

LOGISTIC REGRESSION — Titanic Survival Prediction

What is Logistic Regression

Logistic regression is called regression because internally it predicts a continuous probability value between 0 and 1, not a class label directly. It takes a linear combination of input features and passes it through a sigmoid function, which squashes any value into the range 0 to 1. This output is the probability of belonging to the positive class. If that probability crosses 0.5 we predict survived, below 0.5 we predict died.

The sigmoid and cross entropy plot makes this intuitive. The sigmoid curve on the left shows the S shaped function mapping inputs to probabilities with the decision boundary at 0.5. The region to the right is predicted survived and to the left is predicted died. The cross entropy loss chart on the right shows what happens when the model is confidently wrong. If the true label is survived but the predicted probability is close to zero, the loss shoots toward infinity. This forces the model to never be arrogantly wrong, which is what makes cross entropy the right loss function for classification. MSE does not punish confident wrong predictions nearly as harshly, which is why it is not used here.

The loss function measures error for a single passenger. The cost function is the average of all those individual losses over the full training set. During training we minimize the cost function. We divide the gradient by the number of samples to keep gradient scale stable across different dataset sizes.

The decision boundary is found by setting the sigmoid output equal to 0.5. This simplifies to the point where the linear combination of features equals zero. Everything on one side of this boundary gets classified as survived and everything on the other side as died.

Understanding the Data through EDA

Only 38.4% of passengers survived, so we had a class imbalance right from the start. A model that predicted everyone died would still get 61.6% accuracy, which immediately shows why accuracy alone is a misleading metric on imbalanced data. We needed precision, recall, F1, and AUC to properly evaluate performance.

The port, family size and fare analysis chart revealed three key patterns. Passengers from Cherbourg survived at 55.4% compared to 39% from Queenstown and only 33.7% from Southampton. This is likely because more first class passengers boarded at Cherbourg. The family size chart showed medium families of 3 to 4 people had the highest survival rate around 72%, while solo travellers and very large groups both did poorly. The fare distribution plot clearly shows survivors spread toward higher fares while those who died were concentrated at the lower end, directly linking economic status to survival chances.

The cross feature heatmaps gave the sharpest picture of all. First class women survived at 96.8%, second class women at 92.1%, and even third class women at 50%. But men across all

classes did terribly at 36.9% in first class, 15.7% in second, and only 13.5% in third. Gender was the dominant factor in survival, not class alone. The embarkation heatmap showed Cherbourg's advantage was mainly concentrated in first class at 69.4%.

Correlation Matrix

The Titanic correlation matrix showed pclass was negatively correlated with survival at 0.34, meaning lower class numbers (higher class) improved survival odds. Fare had a positive correlation of 0.26 with survival. Family size showed high correlations of 0.89 with sibsp and 0.78 with parch since we created it by combining those two columns, which confirmed the feature engineering was valid. Fare and pclass had a strong negative correlation of 0.55 since higher class passengers paid more.

Feature Engineering and Preprocessing

We created a family size feature by adding sibsp and parch and adding 1 for the passenger themselves. We log transformed fare since it was heavily right skewed with many outliers. Missing age values were filled using the median since age had outliers and median is more robust than mean in that case. For missing fare values we used the median as well. Gender and port were encoded numerically since models cannot process raw text. We used StandardScaler to bring all features to a similar scale before training. Standardization is preferred here over normalization because linear models assume features are on comparable scales and standardization handles outliers better than min max scaling.

Gradient Descent from Scratch

We built logistic regression manually from scratch using gradient descent and binary cross entropy loss. The gradient descent convergence plot shows the loss dropping sharply from 0.7 in the first 250 iterations and flattening around 0.42 after that. The accuracy curve mirrors this, jumping from around 50% to 80% in the same early window and then stabilizing. This from scratch model achieved 80.45% test accuracy, exactly matching sklearn, which confirmed the implementation was correct.

Regularization

Regularization is used to prevent overfitting by adding a penalty to large coefficient values, forcing the model to stay simpler and generalize better to unseen data. It reduces variance in the model. Even though it slightly increases bias, the overall generalization improves.

The regularization analysis chart shows how L1 and L2 behave across different values of the C parameter, which is the inverse of regularization strength. At very low C (very strong regularization) accuracy drops because the model is too restricted. At moderate C both stabilize near 80% test accuracy. The L1 coefficient path on the right shows features disappearing one by one as regularization strengthens. The port features and the alone indicator vanish first. Gender and class hold on the longest, confirming they are the strongest predictors. L1 performs

feature selection. L2 keeps all features but shrinks their influence, and it handles correlated features better.

Evaluation

The ROC curves chart tested four model configurations. All four produced an AUC of around 0.839 to 0.840, showing that regularization type and strength had very little impact on overall discrimination ability. An AUC of 0.839 means the model correctly ranks a random survivor above a random non survivor 84% of the time. All four curves sat well above the diagonal random baseline.

The best model evaluation chart for L2 with C equal to 1 shows three things together. The confusion matrix shows 95 correct deaths and 49 correct survivals with 15 false survival predictions and 20 missed survivors. Precision for the died class was 0.83 with recall 0.86. Precision for survived was 0.77 with recall 0.71, giving an F1 of 0.74. The model is better at detecting deaths than survivals, which reflects the class imbalance.

Precision tells us out of all the passengers we predicted survived, how many actually did. Recall tells us out of all passengers who actually survived, how many did we correctly identify. Precision matters when false alarms are costly. Recall matters when missing a positive case is dangerous. F1 balances both. The precision recall curve showed an average precision of 0.793.

Accuracy is not a good standalone metric here because predicting everyone died gives 61.6% accuracy without learning anything. Our 80.45% is genuinely better but we verify it through precision, recall, F1, and AUC together.

Feature Importance and Odds Ratios

The feature coefficients and odds ratios chart is the most interpretable output of the project. An odds ratio tells you how much each feature multiplies the odds of survival. Being male had an odds ratio of 0.29, meaning male passengers had less than one third the odds of female passengers. Passenger class had an odds ratio of 0.48, so each step down in class nearly halved survival odds. Being alone reduced odds to 0.80. Log fare had an odds ratio of 1.43, meaning higher paying passengers were 43% more likely to survive, capturing both the class and wealth advantage in a single number. These odds ratios connect the mathematical model back to real world meaning in a way raw coefficients do not.

Probability Calibration

The probability calibration chart shows two things. The left panel shows the predicted probability distributions for each class. Pink bars for passengers who died are concentrated near low probabilities while green bars for survivors cluster toward higher probabilities, with the decision boundary at 0.5 as a dashed line. The right panel is a reliability diagram checking whether predicted probabilities are trustworthy. A perfectly calibrated model follows the dashed diagonal exactly. Our model tracks it reasonably with some wobble in the middle range, meaning the

probabilities are useful but not perfectly calibrated. In high stakes applications we would apply Platt scaling or isotonic regression to improve calibration.

Class Imbalance

The dataset had 62% deaths and 38% survivals. Handling this required using stratified train test splits so both classes were proportionally represented. We evaluated using precision, recall, and AUC rather than accuracy alone. More advanced techniques for imbalanced data include SMOTE, which generates synthetic minority samples by interpolating between existing nearby points rather than simply duplicating them. This helps the model learn better decision boundaries. Class weighted loss is another approach that penalizes the model more for getting the minority class wrong.

Bias and Variance

Bias is the error from the model being too simple and missing real patterns, leading to underfitting. Variance is the error from the model being too sensitive to training data, memorizing noise rather than genuine patterns, leading to overfitting. You cannot eliminate both at the same time. Reducing bias by making the model more complex tends to increase variance and vice versa. Regularization directly controls variance by penalizing complexity. Our logistic model had a 5 fold cross validation accuracy of 79.91% with a standard deviation of only 1.05% and a test accuracy of 80.45%, showing low variance and a healthy generalization gap.

Final Results

The best logistic regression model (L2, C equal to 1) achieved 80.45% test accuracy, AUC of 0.839, and average precision of 0.793. The 5 fold cross validation confirmed 79.91% accuracy with a standard deviation of 1.05%, showing the model is stable across different data splits. Gender was the single most powerful predictor followed by passenger class and fare. Building the model from scratch before using sklearn made the optimization process tangible and confirmed a genuine understanding of how logistic regression finds its weights through gradient descent on cross entropy loss.